Moral Reasoning on the Pareto Frontier

A multi-objective view of epistemic and ethical values, and why equilibrium-style methods are well-suited to traverse it

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Abstract

What counts as good moral reasoning when there is no single ground truth? We propose a multi-objective view in which a moral position — a set of commitments together with a set of principles that account for them — is evaluated against two axis families that genuinely conflict. The primary axes are *ethical*: principles carry fuzzy loadings on moral values, and we work with five — FREEDOM, SOLIDARITY, JUSTICE, WELFARE, and DIGNITY — chosen so that the central tensions of moral life (liberty vs. community, fairness vs. welfare, welfare vs. dignity) appear directly as trade-offs among axes. The secondary axes are *epistemic* measures of internal coherence, drawn from the formal account of Beisbart et al. (2021): account, systematicity, and faithfulness. Every reachable position occupies a point in this joint space, and the rational object of study is its *Pareto frontier*: the set of attainable positions for which no other position is at least as good on every axis and strictly better on at least one. We argue that this reframes the central question of moral reasoning from "which position is right?" to "which strategy reliably finds positions on the frontier?" — and that the question favours *equilibrium-style methods* that balance competing values, because without a ground truth the best a reasoner can do is settle on a defensible middle. We sketch a pilot using MoReBench dilemmas and an LLM-based reasoning system we have already prototyped.

1 Motivation: moral reasoning without a ground truth

Moral reasoning has long resisted the model of justification that works elsewhere in epistemology. There is no external fact of the matter that a moral position *tracks* the way an empirical belief tracks the world. Two careful reasoners working from different starting intuitions can both reason responsibly and reach different conclusions, and we do not, on reflection, want to say one of them is just wrong — pluralism in moral epistemology is widely accepted (Elgin, 1996; Rechnitzer, 2022). The natural question, then, is not which position is correct but *what counts as good reasoning* in a domain where the standard correctness criterion does not apply.

We argue that this question has a structural answer: good moral reasoning is *multi-objective*. A moral position is graded on at least two distinct axis families. The primary ones are *ethical*: a position carries weight on different moral values — freedom, solidarity, justice, welfare, dignity — and these values genuinely conflict. A position that maximises welfare may underweight freedom; a position centred on dignity may not maximise welfare; a position that elevates solidarity may sit uneasily with strong protections of individual freedom. The choice of how to balance such tensions is what moral reasoning is for.

In parallel, a position is graded *epistemically* on how well it hangs together as reasoning — the principles should account for the commitments, the principles should be compact and unifying, and the commitments should not stray too far from the agent's starting intuitions. A formal account of these coherence properties is available (Beisbart et al., 2021). We take these as secondary objectives: a morally interesting position that is internally incoherent is not a serious candidate, but a perfectly coherent position can still occupy any number of places in value-space, and the central philosophical interest is in the latter.

When objectives genuinely conflict, the rational object of study is not a single optimum but a *Pareto frontier*: the set of attainable positions for which no other position is at least as good on every objective and strictly better on at least one. We claim that this is the right object of study for moral reasoning, and that it gives us a new way to evaluate reasoning strategies. A good strategy is one that reliably finds Pareto-optimal positions — and ideally finds interior (mixed) ones, since corner solutions ("maximise welfare and nothing else") are easy and uninteresting. We will argue that *equilibrium-style* methods — methods that proceed by mutually adjusting commitments and principles in light of each other — are particularly well-suited to this role, because they explicitly negotiate the kind of trade-offs the frontier consists of. Reflective equilibrium, in this view, is one instance of a broader family of equilibrium-style traversal strategies worth studying empirically.

2 The moral data resource

The methodological contribution we are pitching is a way to make moral reasoning empirically studyable: construct a curated, mathematically defined *moral data resource*, then study reasoning strategies as pure traversal algorithms on it. Three ideas combine.

Pre-enumerated inputs. The candidates an agent might consider when reasoning about the dataset — specific case verdicts and candidate moral principles — are pre-enumerated. We construct a finite pool of candidate commitments and a finite pool of candidate principles from a dilemma source, then curate both pools with explicit quality control: fusion of near-duplicates, deletion of weak candidates, per-principle scoring on quality dimensions (independent systematicity, specificity, domain-fit), and philosopher review of every retained candidate. The curated pool is intended as a reusable resource in its own right, independent of any downstream reasoning method.

Pre-enumerated inferential connections. For every principle-commitment pair we pre-compute whether the principle entails the commitment, contradicts it, or is silent on it. The result is a fixed, philosopher-validated inferential graph between principles and commitments.

Pre-tagged ethical values. Each principle in the curated pool carries a fuzzy value-loading vector distributing its weight across the five values, also fixed during curation.

Reasoning becomes pure mathematics. The combined output is a fully specified moral space. *Reasoning over it is then a purely mathematical operation: a reasoning strategy is simply an algorithm that selects, from the fixed pools, a subset of commitments and a subset of principles subject to consistency, graded by the multi-objective evaluation defined in Section 3. No LLM participates in the reasoning loop.* The LLM does its work during construction; all comparison of strategies happens on a fixed substrate.

This decoupling is what makes the comparison cleanly interpretable. Current debates about LLMs and moral reasoning conflate the *moral content* a model can articulate with the *reasoning strategy* it employs. Here those are separated: the moral content lives in the curated pool, validated up-front; the strategies are pure mathematics operating on it.

3 A formal model of the moral solution space

We adopt a formal model adapted from Beisbart et al. (2021) and the simulation infrastructure of Freivogel (2023), extended with an explicit value layer. We summarise the notation here and then state the multi-objective optimization problem it gives rise to.

Pools. Fix a finite pool of candidate moral commitments $\mathcal{S}$ — case verdicts indexed by the dilemmas in our dataset, closed under negation — a finite pool of candidate moral principles $\mathcal{P}$, and a finite set of moral values $\mathcal{V} = \{v_1, \dots, v_K\}$. In this work $\mathcal{V} = \{\textsc{freedom}, \textsc{solidarity}, \textsc{justice}, \textsc{welfare}, \textsc{dignity}\}$. Note that $\mathcal{S}$ is not a domain-general pool of moral propositions; it is data-set-specific.

Value loading. Each principle carries a fuzzy value-loading vector

$$\boldsymbol{v} : \mathcal{P} \to [0,1]^K, \qquad \textstyle\sum_k \boldsymbol{v}_k(p) = 1.$$

Values are properties of principles, not of commitments, on the view that commitments are case verdicts ("in dilemma d, action a is required") whereas principles articulate *why*, and the *why* is what carries value content.

Dialectical structure. The pools are linked by a relation $\rho : \mathcal{P} \times \mathcal{S} \to \{\text{entails}, \text{contradicts}, \text{silent}\}$ that records, for every principle–commitment pair, the inferential relation between them. The triple $\tau = \langle \mathcal{S}, \mathcal{P}, \rho \rangle$ plays the role of Beisbart's dialectical structure and defines the search space.

Epistemic state. An *epistemic state* is a pair $\sigma = (C, T)$ with $C \subseteq \mathcal{S}$ minimally consistent (never contains both s and $\neg s$) and $T \subseteq \mathcal{P}$. The initial commitments $C_0 \subseteq \mathcal{S}$ are fixed before reasoning begins and serve as a faithfulness anchor.

Objectives. A state σ is evaluated by K ethical-value measures and three epistemic measures:

$$\begin{aligned} V_k(\sigma) &= \tfrac{1}{|T|} \textstyle\sum_{p \in T} \boldsymbol{v}_k(p) \in [0,1], \quad k = 1, \dots, K \\ A(\sigma) &= \text{degree to which } T \text{ accounts for } C \in [0,1] \\ S(\sigma) &= \text{systematicity of } T \in [0,1] \\ F(\sigma \mid C_0) &= \text{faithfulness of } C \text{ to } C_0 \in [0,1] \end{aligned}$$

The value measures aggregate the loadings of the principles in T. The three coherence measures are operationalised following Beisbart et al. (2021); we refer to their paper for definitions. Together they give a *multi-objective optimization problem* over Σ, the set of admissible epistemic states:

$$\max_{\sigma \in \Sigma} \big(V_1(\sigma), \dots, V_K(\sigma),\, A(\sigma),\, S(\sigma),\, F(\sigma \mid C_0) \big).$$

Pareto dominance and frontier. A state σ' *Pareto dominates* σ, written $\sigma \prec \sigma'$, if every objective at σ' is at least as large as at σ and at least one is strictly larger. The *Pareto frontier* of the problem is

$$\Pi(\tau, C_0) \;=\; \big\{ \sigma \in \Sigma : \nexists\, \sigma' \in \Sigma \text{ with } \sigma \prec \sigma' \big\}.$$

Π is the empirical object this paper proposes to study.

Implementation status. As a precursor to the curated-pool pilot of Section 7, we have built an end-to-end *LLM-in-loop* prototype of this formal setup (six agents: a commitment former, two inferencers, three verifiers) on the five trolley cases of Rechnitzer (2022). This prototype is transitional — it puts an LLM inside the reasoning loop because the curated pool described in Section 2 does not yet exist for the dataset — but it has already produced informative cross-model variance. Three small-model runs (Gemini 2.0 Flash, Claude 3.5 Haiku, GPT-4o-mini) land at qualitatively distinct epistemic states (a clean doing-vs-allowing theory; a consequentialist reformulation with conditional carve-outs; an under-abstracted enumeration of the cases), already suggesting that *which point of* Π *is reached* is doing more work than *whether* a point of Π is reached. The proposed pilot pulls the LLM out of the loop and runs the comparison on a fixed, philosopher-validated τ.

4 Two axis families, one frontier

The objective vector splits naturally into two axis families. The primary $(V_1, \ldots, V_K) \in [0,1]^K$ is *ethical* — properties of the position as a moral stance. The secondary $(A, S, F) \in [0,1]^3$ is *epistemic* — properties of the position as reasoning. The two families are formally on the same footing inside the optimization problem above but answer different questions and trade off against each other in characteristic ways.

Why values, not frameworks. We deliberately do not use named ethical frameworks (utilitarianism, Kantianism, virtue ethics) as the value axes. Frameworks are bundles: Kantian deontology loads on DIGNITY *and* on a particular kind of JUSTICE; utilitarianism loads on WELFARE *and* a particular kind of SOLIDARITY. Values are more fine-grained, and the moral tensions that drive reasoning — FREEDOM against SOLIDARITY, WELFARE against DIGNITY — are most clearly articulated at the value level. Frameworks can then be recovered as characteristic value mixes *a posteriori*, but the reverse is lossy.

Two frontiers worth studying. Projecting $\Pi(\tau, C_0)$ onto the epistemic subspace and the ethical subspace gives two lower-dimensional frontiers, Π_E and Π_V respectively. They are related but not collapsible: an epistemic Pareto-optimal state need not be ethical Pareto-optimal (it may achieve high coherence at a corner of value-space) and vice versa. A reasoning strategy that consistently lands on $\Pi_E \cap \Pi_V$ — a position that is coherent *and* pareto-optimal across values — is doing something philosophically distinctive, and it is this intersection that we propose as the target of "good moral reasoning".

[Concept diagram — generate with Nano Banana, prompt in `figures/nano_banana_prompt.txt`]

Figure 1: **Building the moral data resource.** Source dilemmas → LLM generation of commitment and principle pools → *curation* (fuse, prune, score per-principle systematicity; philosopher review) → value tagging $\boldsymbol{v} : \mathcal{P} \to [0,1]^K$ → dialectical structure $\tau = \langle \mathcal{S}, \mathcal{P}, \rho \rangle$. The output is a curated, mathematically defined moral space over which any traversal method can be evaluated.

5 Two frontiers, one trajectory

Figures 2 and 3 show the two projected Pareto frontiers Π_E and Π_V as we expect them to look once the pilot is run. The first projects Π_E onto (A, S) with F held near 1 (an agent who hasn't drifted far from C_0); the second projects Π_V onto two representative value axes.

In each figure we sketch the trajectory of a single RE run as a sequence of *accepted* and *rejected* adjustment proposals. The pattern in the epistemic plot is familiar from our trolley pilot: an A-step lifts A at modest S cost, then a B-step expands C to admit an emerging commitment (briefly hurting A), and then the next A-step articulates a sharper principle that lifts both axes. Rejected moves — the brittle ones that hill-climbing rules out — map roughly to dominated points off the frontier.

The value-space trajectory shows something different and more interesting. The same alternating B/A pattern, projected onto $(V_{\text{freedom}}, V_{\text{solidarity}})$, traces a path that ends not at a corner (pure freedom or pure solidarity) but at an interior point of the frontier — a value mix that neither value alone would predict but that no other mix dominates either. This is what "reaching an equilibrium" looks like in value-space, and we conjecture it is the real philosophical contribution of RE: a method that finds defensible *mixed* positions in a domain where corner solutions are easy and interior solutions hard.

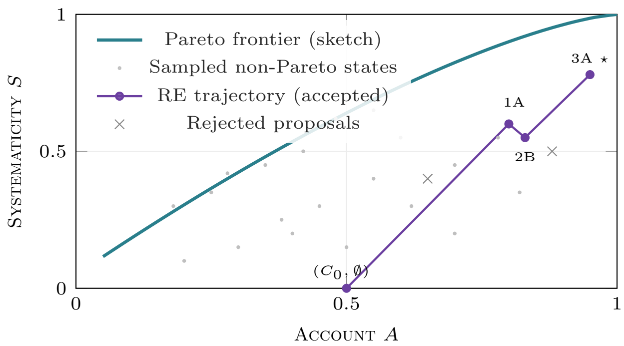

Figure 2: **Epistemic Pareto frontier.** Projection onto (A, S) with $F \approx 1$. The RE run climbs through three accepted moves and rejects two dominated proposals (grey $\times$). The starred final state sits at the frontier — a global trade-off optimum no other reachable state dominates.

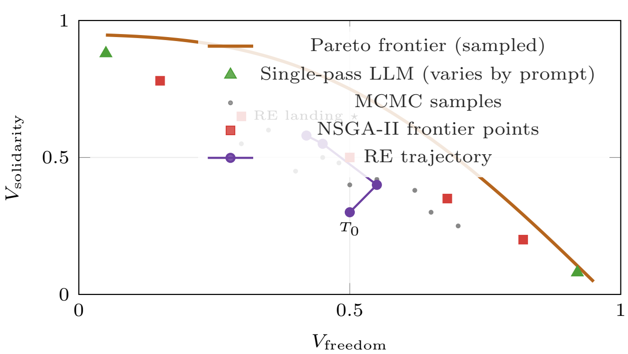

Figure 3: **Ethical-value Pareto frontier.** Projection of $V(T)$ onto $(V_{\text{freedom}}, V_{\text{solidarity}})$. Single-pass LLM judgment collapses to corners depending on prompt. NSGA-II traces the frontier. RE lands at an interior frontier point — a defensibly mixed normative profile.

6 Hypotheses

We propose two empirically testable hypotheses about reflective equilibrium.

H1 (Interior-bias). RE lands on the *interior* (mixed) region of the ethical-value Pareto frontier Π_V more reliably than do non-equilibrium baselines. The alternation between commitment and principle revision penalises both under-coverage and over-specialisation, biasing the trajectory toward defensible value mixes rather than monomaniacal corners. H1 articulates what we mean by good moral reasoning in this framework.

H2 (Systematicity-driven value bias). The systematicity desideratum biases RE toward values that admit compact principle-level statement (WELFARE, JUSTICE) and away from those that resist it (SOLIDARITY, DIGNITY). If confirmed, this is a substantive philosophical finding: epistemic coherence preferences silently shape the values a moral reasoner ends up endorsing.

7 Methodology sketch

Data. We use MoReBench (Scale Labs, 2025) and in particular the MoReBench-Theory subset (150 scenarios with explicit framework labels), which gives us framework-tagged dilemmas without doing the tagging ourselves.

Pipeline. The construction of the data resource is a careful multi-stage process; the resource itself is what makes the downstream traversal comparisons cleanly interpretable.

(i) *Generation.* For each of ∼10 dilemmas, LLM-generate a commitment pool (∼30 candidates) and a principle pool (∼20 candidates: half value-archetypal, half bottom-up from the commitments).

(ii) *Curation.* This is the load-bearing stage. We fuse near-duplicate principles, drop principles with weak independent credibility, and score each surviving principle on quality dimensions independent of the equilibration loop — per-principle systematicity (how general / unifying is this principle taken on its own?), specificity, and domain-appropriateness. The curated pool is then double-checked by philosophers, both to verify the value tagging in (iii) and to catch principles whose moral content the LLM has misread. The intent is that the final pool is a high-quality *moral resource* in its own right — something a researcher could reuse independently of any particular traversal method.

(iii) *Value tagging.* LLM-tag each surviving principle with a value-loading vector $\boldsymbol{v}(p) \in [0,1]^K$ over the five values; MoReBench-Theory's framework labels serve as calibration anchors (utilitarian-framework labels seed welfare loadings, Kantian-framework labels seed dignity / justice, etc.). Philosopher review confirms loadings.

(iv) *Dialectical structure.* Pre-compute $\rho : \mathcal{P} \times \mathcal{S} \to \{\text{entails}, \text{contradicts}, \text{silent}\}$ for every principle-commitment pair as a one-time batch.

The output of this pipeline is a *curated, mathematically well-defined moral space* $\tau = \langle \mathcal{S}, \mathcal{P}, \rho \rangle$ together with the value-loading map $\boldsymbol{v}$, on which any traversal method can be evaluated.

Traversals. The primary object of study is symbolic RE on the curated pool. To put its behaviour in context we run a small set of comparison traversals on the same pool: a single-pass LLM judgment baseline (with no equilibration), and a NSGA-II reference that estimates Π_V and Π_E without committing to any particular reasoning strategy. Each yields a distribution of $(A, S, F, V_1, \ldots, V_K)$ points.

Analysis. Locate RE's landing distribution on Π_E and Π_V relative to NSGA-II and the single-pass baseline. Test H1 (interior bias) by Mann-Whitney displacement against single-pass on Π_V; test H2 (systematicity-driven value bias) by directional displacement of RE landings relative to NSGA-II frontier points at matched epistemic cost.

8 Discussion

The framing answers two long-standing complaints about RE without rejecting them. *Pluralism* (de Maagt, Singer) is real but not relativism: the frontier picks out a proper subset of all positions, and being on it is a substantive constraint. *Conservatism* (Brandt, Kelly–McGrath) is real but not damning: it shows up as a systematic landing-region offset toward $F = 1$, which a method comparison can quantify rather than dismiss. And the underlying philosophical question — what *is* good moral reasoning? — becomes empirically tractable as a property of traversal strategies relative to a measurable solution topology.

The contribution we are aiming at is not another moral-reasoning benchmark, nor another RE-vs-critics dialectic. It is a measurement instrument: value-trajectory plots and Pareto-region analyses that any LLM-based or symbolic moral reasoning system can be characterised against.

References

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